

Can We Reconstruct a Diagnostic 12-Lead ECG from Only a Few Measured Leads?

Discrete Latent Sequence Generation for Limited-Lead ECG Reconstruction

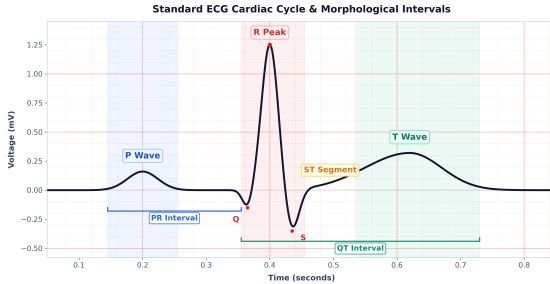
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12 Leads vs. One Wearable Lead



Electrophysiological Fiducials

P wave (atria), QRS (ventricles), and ST-T
(repolarization)

Clinical 12-Lead ECG

12 Spatial Views

- Frontal (I-aVF) + Precordial (V_1 - V_6)
- Comprehensive 3D cardiac vector field

Smartwatch ECG

1 Measured Projection

- Single wrist transverse vector (Lead I)
- Completely blind to vertical & chest planes

Most spatial information is unobserved.

The Reconstruction Challenge

One Observed Lead



?



12-Lead ECG

Spatial Ambiguity

Different cardiac electrical states can produce similar single-lead measurements.

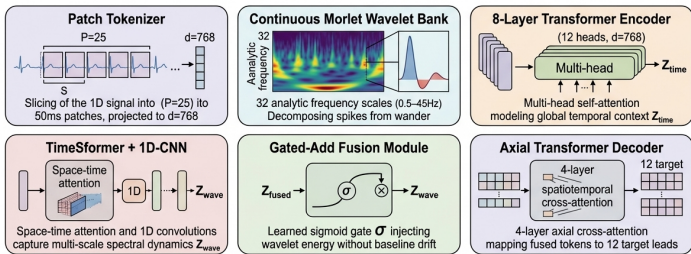
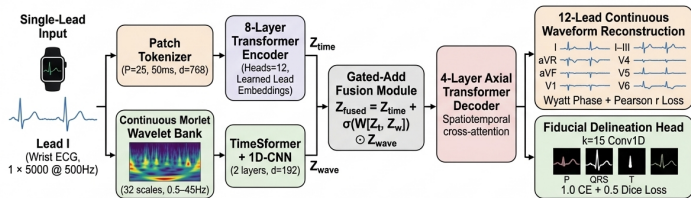
Generative Inference

The model therefore has to infer plausible missing morphology without simply smoothing it away.

ECG-AIM Model Architecture

ECG-AIM Architecture (conv15e_A0_wave_noSSL_gated_add)

Single-Lead Time-Frequency Wavelet Conditioning & Multi-Task Clinical Supervision



Experimental Setup

~175k ECGs

Trained and benchmarked across diverse cohorts from **PTB-XL** and **MIMIC-IV-ECG**

Competitive Baselines

Direct head-to-head comparison against continuous **1D U-Net** and **MultiScale-VAE**

1 to 3 Observed Leads

Evaluated from single wearable Lead I up to 3 complementary leads (I, II, V_2)

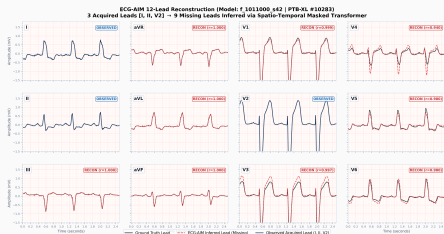
Comprehensive Evaluation

Evaluated on **waveform correlation**, **QRS timing**, and **downstream clinical preservation**

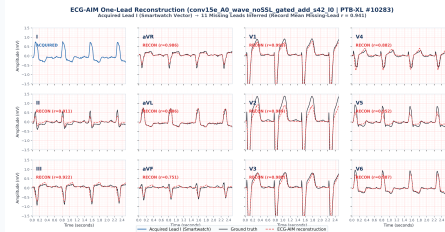
Spatial Information Determines Reconstruction Quality

3 Leads Acquired (I, II, V_2):

$$r > 0.98$$



1 Lead Acquired (Smartwatch Lead I): $r = 0.776$



Adding complementary spatial measurements sharply improves reconstruction. · Spatially complementary leads provide substantially more information for reconstruction.

Preservation of Clinically Relevant Structure

Morphology

0.776

Precordial Lead r
(vs. 0.571 U-Net, 0.684 MS-VAE)

Timing

8.0 ms

QRS Timing Error
(Meets < 10.0 ms Clinical Gate)

Downstream Task

0.775

EchoNext Structural AUROC
(Ground Truth = 0.803)

The reconstruction preserved more than waveform similarity.

Conclusions and Translational Next Steps

1. Reconstruction is limited by the spatial information observed. Single wearable Lead I recovers useful morphology ($r = 0.78$); 3 complementary leads unlock near-lossless recovery ($r > 0.98$).

2. ECG-AIM preserves substantial morphology and timing from limited leads. Reconstructed signals meet clinical timing gates (8.0 ms QRS error) and retain downstream diagnostic utility (0.775 EchoNext AUROC).

3. Prospective paired smartwatch-12-lead validation is the next test. Translating from retrospective archives to live clinical deployment requires validating across live hardware factors.

Sunnybrook paired validation protocol in preparation · PI: Dr. Christopher Cheung
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